

CampusShare

Technical Stack Decision Document

Project: CampusShare — A verified student marketplace for buying, selling and renting academic resources.

1. Project Overview

CampusShare is a campus-focused marketplace that helps students discover, buy, sell and rent academic resources. The platform combines marketplace listings, image-based product information, student authentication, wishlist functionality and buyer–seller communication.

2. Technology Stack

Layer	Technology	Purpose
Frontend	React.js + Vite	Component-based UI and production builds
UI	Tailwind CSS + Lucide React	Responsive styling and consistent icons
Routing	React Router DOM	Client-side navigation and protected routes
Server State	TanStack React Query	API fetching, caching and synchronization
Backend	Node.js + Express.js	REST API and application server
Database	MongoDB Atlas + Mongoose	Cloud document database and data modeling
Authentication	Firebase Authentication + Google Sign-In	User authentication and Google OAuth
Server Auth	Firebase Admin SDK	Server-side identity-token verification
Uploads	Multer	Multipart/form-data and image uploads
Deployment	Vercel + Render	Frontend and backend hosting
Version Control	Git + GitHub	Source control and deployment workflow

3. Why This Stack Was Chosen

- **React + Vite:** component-driven frontend suitable for a multi-screen marketplace.
- **Node.js + Express:** straightforward REST API architecture using JavaScript across the application.
- **MongoDB Atlas:** flexible document modeling for listings, profiles, categories and conversations with managed cloud hosting.
- **Firebase Authentication:** reliable Google Sign-In without implementing OAuth from scratch.
- **Mongoose:** schema definitions, validation, references and indexes for MongoDB.
- **Vercel + Render:** simple separation of frontend and backend production deployment.

4. Core Features in Scope

- Google-authenticated student accounts
- Marketplace browsing and category discovery
- Buy, sell and rent listing workflows
- Listing images and product information

- Indian Rupee (₹) pricing and rental pricing
- Wishlist functionality
- Seller/buyer conversations and messaging
- Protected routes and profile/account management

5. Features Kept Out of Scope

The current implementation does not claim a full payment gateway, escrow system, delivery logistics or production transaction settlement. These are future extensions.

6. Architecture

React/Vite client hosted on Vercel → **Express/Node.js API** hosted on Render → **MongoDB Atlas**. Firebase Authentication handles Google Sign-In and the backend verifies authenticated identity for protected operations.

7. Database Design

The accompanying DBML file documents the logical collections, fields and references for users, categories, listings, wishlists, conversations and messages. It can be imported into dbdiagram.io for visualization.

8. Submission Note

This document accompanies the CampusShare source-code repository and database schema for the Technical Internship — Project Development Task.